

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 1
Subject: 031011321 Mechanical Drawing I
Date: Friday, August 8, 2025
Instructor: Mr. Inthrayuth Mahaphol (INMAH), Mr. Pichai Chobsuk (PICHO)

Year: 2025
Section: 8
Time: 13.00-15.00

Name: _____ ID: _____ Class: _____ No: _____

Directions:

1. This examination paper has not been edited. If any errors are identified, students are advised to exercise discretion in addressing them.
2. The examination consists of 6 questions on 6 pages (including the cover page).
The total score is 36 points.
3. Complete all questions in the answer booklet.
4. Books, documents and lecture notes are not allowed
5. Electronic calculators are not allowed
6. Students must remain in the examination room for no less than 1 hour from the start of the examination.
7. Do not open or start the examination before permission is granted. Students must strictly follow the instructions of the examination.
8. Access to restroom facilities during the examination is not permitted, except in cases of emergency.
9. Students are not allowed to take the examination paper or copy the examination paper out of the examination room. Otherwise, it will be considered cheating.
10. Usage of all communication devices during the examination is strictly prohibited.
11. Smart watches (electronic watches) are not permitted during the exam.
12. Allow use of drawing devices and do not borrow equipment during the exam.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

This exam paper is part of a set used to assess student abilities as follows:		Q1	Q2	Q3	Q4	Q5	Q6
CLO 1	Can tell the standards for mechanical drawing.	✓	-	-	-	-	-
CLO 2	Apply applied geometric drawing with drafting equipment.	-	✓	-	-	-	-
CLO 3	Able to draw system projections, Angle 1, and Angle 3 with drawing equipment.	-	-	✓	✓		✓
CLO 4	Write 3D images and sketches with styling tools	-	-	-	-	✓	-
CLO 5	Can be applied to determine the size of the work piece.	-	-	-	-	-	-
CLO 6	Draw and read straight, oblique, pyramid, cylindrical, conical, and circular projection images with drawing equipment.	-	-	-	✓	✓	✓
CLO 7	Write cut images and special cut images. With a drafting device.	-	-	-	-	-	-

1. Answer all questions by circle the number. Each question has only one correct answer. (4 points)

031011321-2

1.1 What is the standard A4 drawing paper (มอก.210-2520)

1. 297 x 420 mm

2. 210 x 297 mm

3. 148 x 210 mm

4. 105 x 148 mm

5. 74 x 105 mm

1.2 How many A4 drawing papers can be fit in an A1 drawing paper?

1. 8 pieces

2. 10 pieces

3. 12 pieces

4. 14 pieces

5. 16 pieces

1.3 Given an actual 120 mm line. How long the line will be drawn in scale 1:5?

1. 20 mm

2. 24 mm

3. 40 mm

4. 60 mm

5. 100 mm

1.4 What is the hidden line of dashed line?

1. 

2. 

3. 

4. 

5. 

1.5 Which are the angles of Isometric?

1. 30° , 30°

2. 30° , 90°

3. 7° , 42°

4. 30° , 47°

5. 0° , 45°

1.6 Which are the angles of Dimetric?

1. 30° , 30°

2. 30° , 90°

3. 7° , 42°

4. 30° , 47°

5. 0° , 45°

1.7 Which are the angles of Oblique?

1. 30° , 30°

2. 30° , 90°

3. 7° , 42°

4. 30° , 47°

5. 0° , 45°

1.8 Which symbol represents the first-angle projection?

1. 

2. 

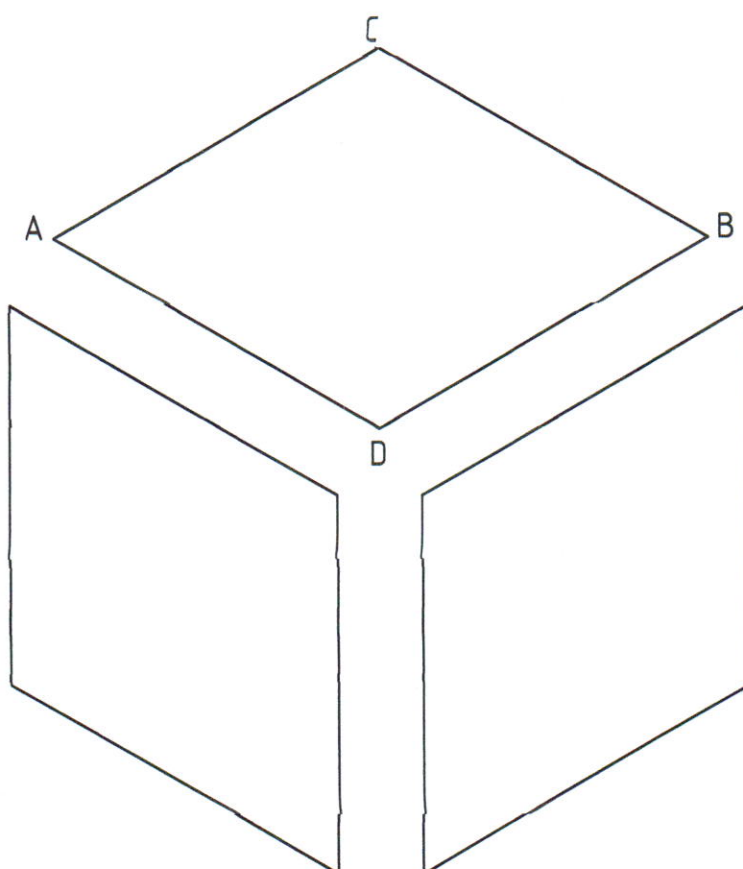
3. 

4. 

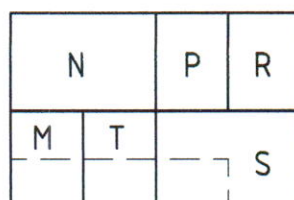
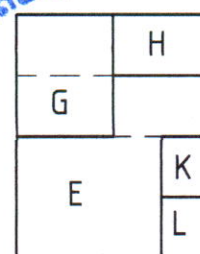
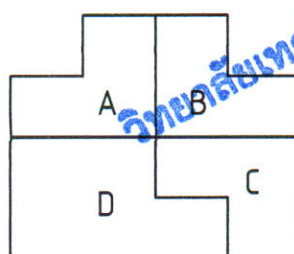
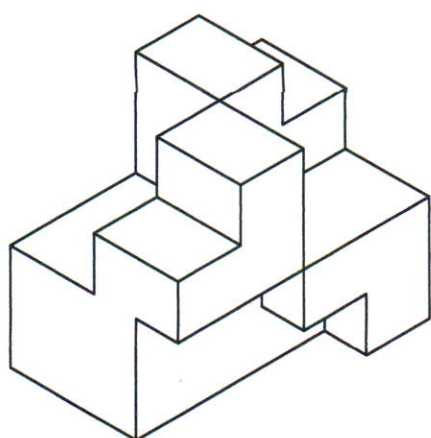
Name..... No..... Section.....

2. Draw ellipses (Isometric) by using the given rhombus (3 points)

031011321-3



3. Answer all the following questions using the given first-angle projection (2 points)



3.1 Which letters indicate the highest surface level?

3.2 What a letter is the same level as R?

3.3 What a letter is the same level as A?

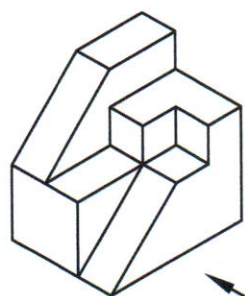
3.4 What a letter is the same level as G?

Name..... No..... Section.....

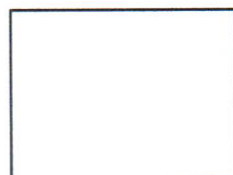
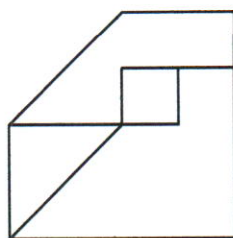
4. Draw three views of the first-angle projection and show projection line (10 points)

031011321-4

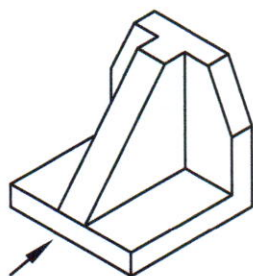
4.1 (2.5 points)



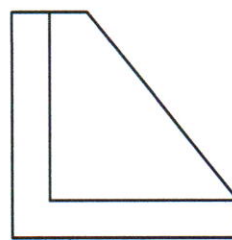
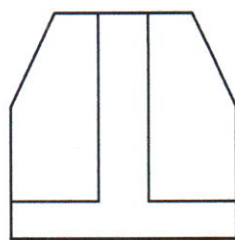
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4.2 (2.5 points)

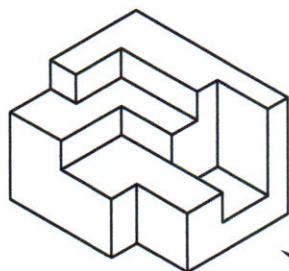


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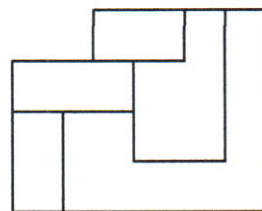


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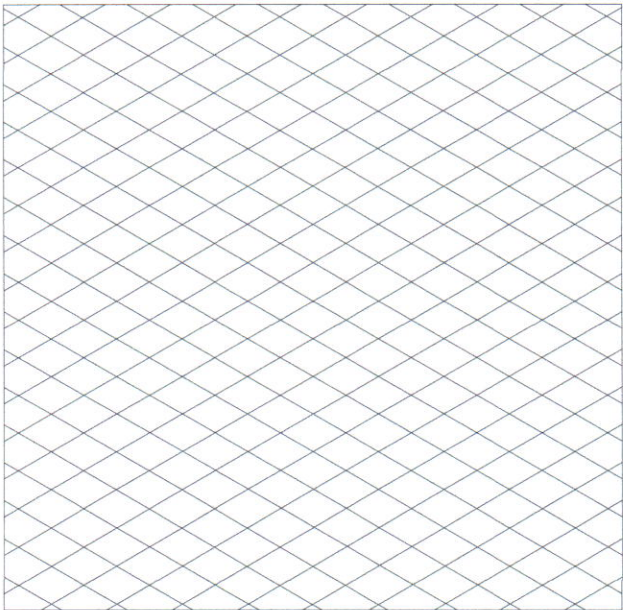
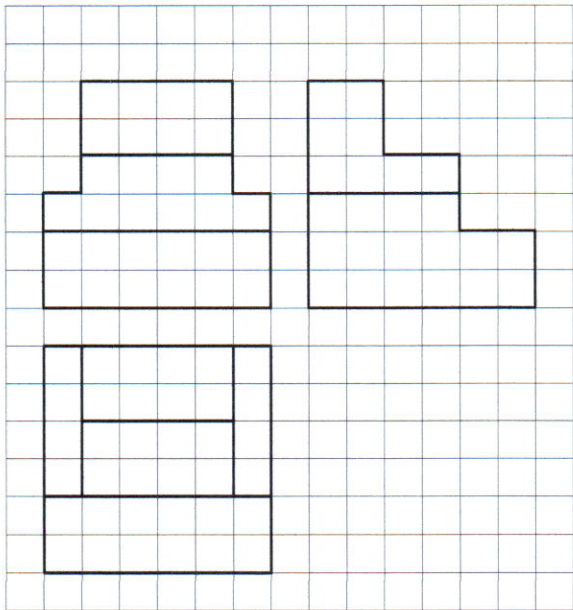
4.3 (5 points)



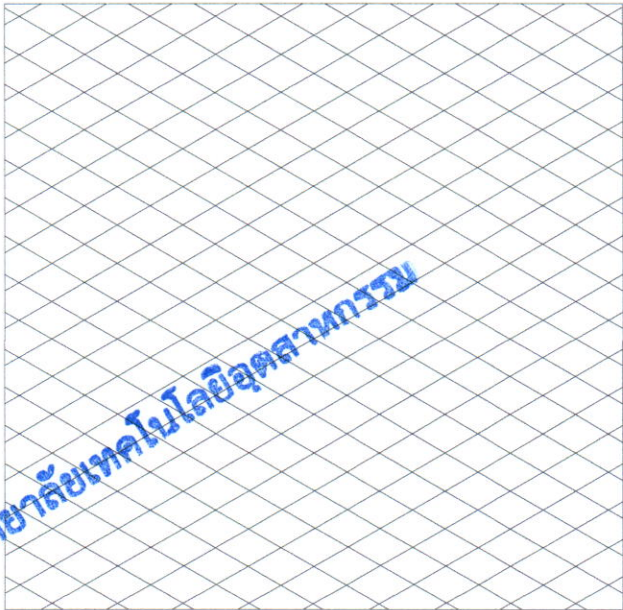
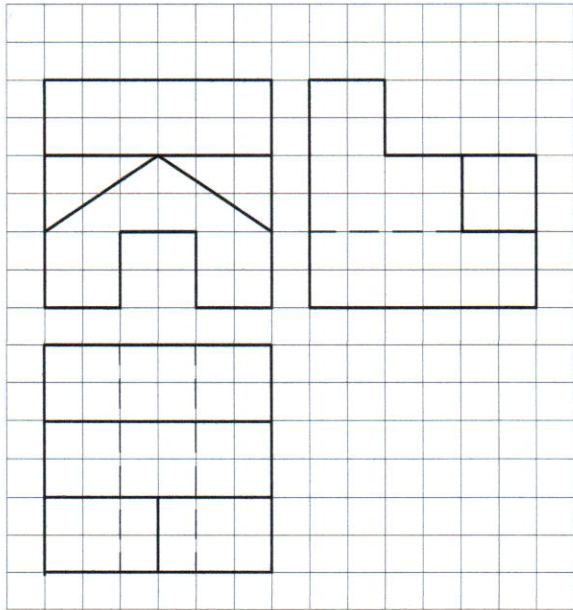
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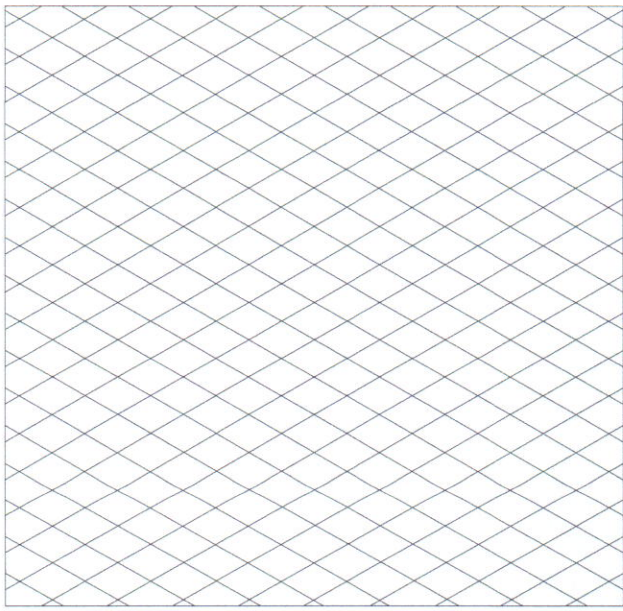
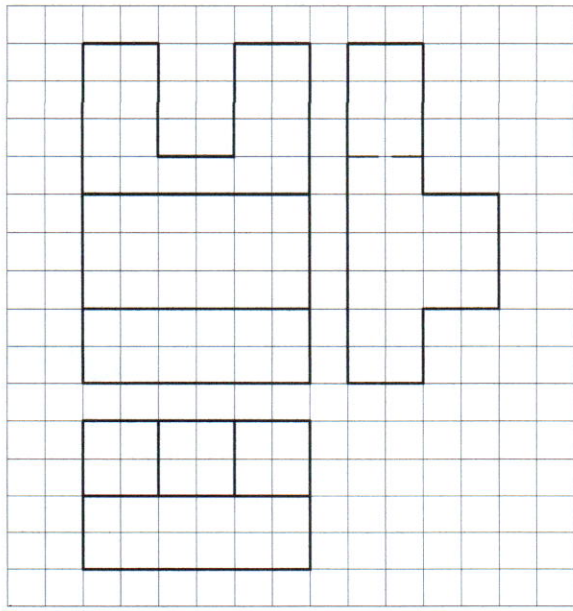
5.1 (3 points)



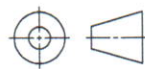
5.2 (3 points)



5.3 (3 points)

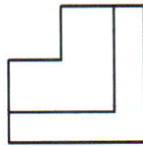
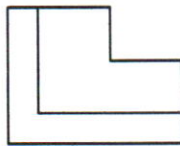
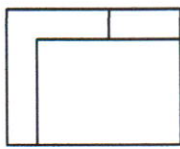
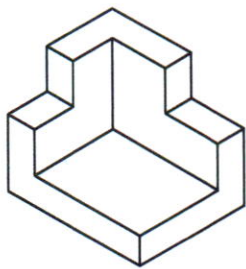


6. Complete the given orthographic projection (4 points)

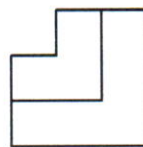
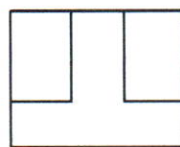
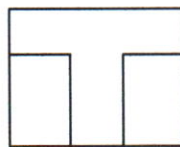
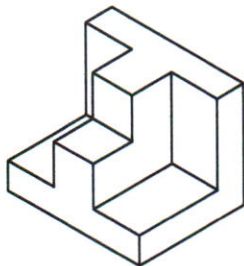


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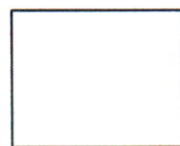
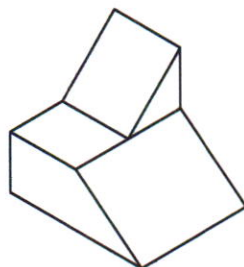
6.1 (1 point)



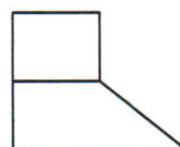
6.2 (1 point)



6.3 (1 point)



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6.4 (1 point)

